Water at Twenty-Four Functions: Slater-Type Bases Set on the Free Atom Against cc-pVDZ, With the Qubit Cost*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

(Dated: October 31, 2026)

Central result. At a fixed basis size of twenty-four functions, the size of cc-pVDZ on water, we ask how much of the Gaussian basis's correlated energy a one-centre Slater-type design recovers when its exponents are set on the free atoms, and what that costs in qubit resources. [**MEASURED**] A Slater double-zeta-plus-polarization basis built by cc-pVDZ's own design rule sits 40.7 mHa below cc-pVDZ at frozen-core CCSD(T), of which 28.8 mHa is self-consistent field. [**MEASURED, negative**] Every exponential kit with fixed or shared exponents loses: the hydrogenic Z/n rule by 196 mHa, a single Coulomb–Sturmian scale per atom by more than one hartree, and a two-scale Sturmian kit by 130 mHa; the hydrogenic loss is the fixed radial node structure, not the decay rate. [**MEASURED, pre-registered**] Releasing the oxygen exponents one at a time, every value set by the free oxygen atom's radial Hartree–Fock energy with nothing tuned on water, closes that gap: with five oxygen exponents the water energy lands 3.1 mHa below cc-pVDZ, and 7.0 mHa below with hydrogen exponents set on H_2. The split of the $2p$ from the $3p$ exponent carries 0.900 of the water gain, as it carries 0.902 of the atom's; the water gain is 0.516 of the atom's. Both were predicted before the water energies were computed. The one non-monotone rung is diagnosed: its correlation loss transfers from the atom one to one and is mostly a $3d$ exponent the atomic objective never sees, while its Hartree–Fock gain is two opposite pieces that cancel on water. [**MEASURED**] The Jordan–Wigner Pauli count is the same for every twenty-four-function basis, 107 382 terms on 46 frozen-core qubits, because in canonical orbitals without degenerate pairs it is fixed by the orbital count and the C_{2v} pattern; the 1-norm rises with accuracy, from 405 to 706 Ha across the kits. *Caps.* One geometry and one method per molecule; the ladder and the native kits are water only. The water-tuned Slater basis stays 34 mHa below the best atom-set rung, and 21.7 mHa below once the $3d$ is also set on the atom's correlated energy (pre-registered; water then lands 19.0 mHa below cc-pVDZ). [**MEASURED, pre-registered**] The matched-accuracy comparison brackets the Gaussian cost from both sides: aug-cc-pVDZ at 41 functions stays 8.0 mHa short of the Slater DZP while already costing 9.2× the Pauli terms and 2.7× the 1-norm on 80 qubits against 46, and the crossing lies between it and cc-pVTZ. [**MEASURED, pre-registered**] The same rule on hydrogen fluoride, a second molecule, repeats both findings: the Slater DZP is 59.5 mHa below cc-pVDZ at nineteen functions with the same Pauli count, and aug-cc-pVDZ is still short at 9.3× the terms. [**MEASURED, pre-registered; STOP**] A ring of six hydrogens, with no core, does not: the Slater DZP is only 3.5 mHa below cc-pVDZ at thirty functions, and one more Gaussian polarization shell at 48 functions already beats it, so the matched-accuracy margin is a property of these two molecules, not of exponential functions. Nothing here is a GeoVac lattice structure: the bases are tuned exponential functions evaluated through Gaussian fits, and the angular sparsity of the one-centre framework does not enter.

I. INTRODUCTION

The atomic sector of the Geometric Vacuum programme is built on exponential radial functions: hydrogenic orbitals with decay Z/n, and Coulomb–Sturmians with one shared scale per centre [1, 2]. Its molecular work has either kept those functions and paid for it in accuracy (the composed architecture, Paper 17), or left them for prolate-spheroidal and explicitly correlated constructions on two centres (Papers 11, 12, 19). Water is the first molecule in the series where the integrals over exponential functions on three centres are evaluated to working precision (Paper 59 and the shared-centre channels of CHANGELOG v5.20.10), and so the first where the question of this paper can be asked cleanly.

The question is narrow on purpose. Fix the basis size at twenty-four functions, which is what cc-pVDZ [3] spends on water. Fix the geometry and the correlation method. Then ask three things. First, does a Slater-type basis built by cc-pVDZ's own design rule beat cc-pVDZ at that size, and by how much at the mean-field and the correlated level separately? Second, do the exponential functions the one-centre framework actually uses, with their fixed or shared exponents, survive the same test? Third, if they do not, how many distinct radial scales does oxygen need, and can those scales be set on the free atom rather than on the molecule?

The third question is the one with content. A basis whose exponents are tuned on the molecule being computed is a fit. A basis whose exponents are set on the free atom and then carried to the molecule tests a transfer.

* Paper 62 in the Geometric Vacuum series

The scale ladder of Sec. V was pre-registered as a transfer test: the predictions were written and committed before any water energy of the new rungs was computed, and the record is reproduced in Sec. V with its disposition.

Two things this paper is not. It is not a GeoVac lattice result. The bases here are tuned Slater-type functions evaluated through Gaussian fits, and the angular (Gaunt) sparsity of the one-centre framework [16] plays no part in any number below; Sec. VI says what the qubit count does and does not depend on. And it is not a claim about the basis-set limit. Everything is measured at one size, one geometry and one method, and the caps are listed in Sec. VIII.

II. SOLVER AND VALIDATION

a. Integrals. All one- and two-electron integrals are evaluated by the McMurchie–Davidson scheme [4] over contracted Cartesian Gaussians, through the project's `noci_engine` module. Gaussian bases are used as published. Slater-type primitives $r^{n-1}e^{-\zeta r}Y_{lm}$ enter as Gaussian fits: one shape is fitted per (l, n) and number of Gaussians, by least squares on the radial function, and then scaled by ζ^2 to the required exponent [5]. Because the shape is fitted once and scaled, the fit error of a given (l, n) is the same at every exponent; this matters for the ladder of Sec. V, where rungs differ only in exponents. Twelve Gaussians per valence primitive and eight per polarization function are used throughout. The fitted basis is a legitimate Gaussian basis with exact integrals, so every Hartree–Fock energy below is a strict upper bound for that fitted basis; the fit's convergence toward the true Slater function is checked in Sec. III.

b. Correlation. The correlated solver is the project's own: restricted Hartree–Fock, spin-orbital coupled-cluster singles and doubles, and the perturbative triples correction [6, 7], with an optional frozen oxygen $1s$. It is in `tests/test_water_ccsd.py`. (An external package was not used because none builds on the host; the solver's independence is also what makes the checks below meaningful.) It is validated where its answer is known:

- it reproduces the published STO-3G water reference of the Crawford programming projects [8] to 2×10^{-12} Ha in the triples correction and 4×10^{-10} Ha in the CCSD correlation energy;
- CCSD equals full configuration interaction for two electrons, to 10^{-9} Ha;
- frozen-core CCSD equals active-space full CI on a case where both are computable;
- nine planted wrong answers (sign flips, dropped diagrams, a wrong denominator) were each caught by the tests;
- in the 34-function Slater triple-zeta set of Stark *et al.* for hydrogen fluoride, at their geometry [18], its frozen-core CCSD(T) correlation energy is -0.25610 Ha (12 Gaussians per function; -0.25607 at 8), against their heat-bath configuration interaction value of -0.2558 Ha from a different code, which integrates the Slater functions directly on a numerical grid. Neither number is full configuration interaction, so the 0.3 mHa difference is agreement, not an error bound. The check covers the correlation energy only, since they report no mean-field energies.

c. Atomic reference. Oxygen exponents are set on the free atom by an exact one-centre radial average-of-configuration Hartree–Fock over Slater primitives, with the Slater F^k and G^k integrals in closed form [9]. The code gives helium's single-zeta energy $-729/256$ Ha exactly, and for neon its energy is the limit that the three-dimensional Gaussian-fit route approaches from above as the number of Gaussians grows (-4.9, -0.20, -0.09 mHa at 8, 12, 16 Gaussians). An earlier open-shell route through fractional-occupation SCF orbitals sat 10.7 mHa above the variational average energy, because its orbitals are not variational for that energy; it was replaced by the radial code and is not used.

d. Geometry and conventions. $R_{\mathrm{OH}} = 1.809$ bohr, $\angle\mathrm{HOH} = 104.5°$. Energies are in hartree. "fc" means the oxygen $1s$ is frozen in the correlation treatment; cc-pVDZ is a valence-correlation basis, so the frozen-core energy is the fair yardstick, and it is the scored quantity throughout. All-electron values are reported alongside where they were computed.

III. EQUAL SIZE, EQUAL DESIGN RULE: SLATER DZP AGAINST CC-PVDZ

The fairness rule is cc-pVDZ's own design, with Slater primitives in place of Gaussian ones. cc-pVDZ on oxygen is a general contraction of atom-optimized primitives to $[3s2p1d]$, with the contracted functions being the atomic $1s$ and $2s$ orbitals plus the most diffuse s primitive, the atomic $2p$ plus the most diffuse p primitive, and one d polarization function. The Slater basis here does the same:

- O: four s primitives $(1s, 1s', 2s, 2s')$ and three p primitives optimized on the atom's average-of-configuration Hartree–Fock energy, contracted to {atomic $1s$, atomic $2s$, most diffuse s} and {atomic $2p$, most diffuse p}. The optimized exponents are $9.317, 6.765 \mid 2.710, 1.694$ for s and $5.719, 2.846, 1.450$ for p, with atom energy -74.768205 Ha.
- H: $1s, 1s'$ optimized on H_2 RHF at $R = 1.4$ bohr: 1.212, 0.627.
- Polarization (O $3d$, H $2p$): one exponent each, optimized on frozen-core CCSD of water, for both bases alike. The cc-pVDZ d and p exponents re-optimized the same way move from 1.185 to 1.312 and from

TABLE I. Water at 24 functions, $R_{OH} = 1.809$ bohr, 104.5°. fc = oxygen $1s$ frozen in the correlation treatment. The cc-pVTZ row (58 functions) is the anchor for the Gaussian ladder. Energies in hartree.

Basis	RHF	CCSD(T) fc	CCSD(T) ae
cc-pVDZ (standard)	-76.026795	-76.241025	-76.243143
cc-pVDZ, pol. re-opt.	-76.027257	-76.242201	-76.244383
Slater DZP (12 G)	-76.055573	-76.281739	-76.285720
Slater DZP (16 G)	-76.055644	-76.281811	-76.285792
cc-pVTZ (58 fns)	-76.057162	-76.332191	

0.727 to 0.605; the Slater optima are O $3d$ 2.756 and H $2p$ 1.579. (A Slater $3d$ is diffuse at a given ζ, with $\langle r\rangle = 7/2\zeta$, so these look large next to Gaussian values.)

Polarization is the one place where water enters the design, and it enters both bases identically. A negative result fixed the contraction rule: an uncontracted $3s2p$ set optimized on the atom spends its second s function on the core (10.03, 8.33, 2.32), leaving the valence s single-zeta. That is why cc-pVDZ uses general contraction, and why the Slater basis does too.

[MEASURED] Table I gives the result. At frozen-core CCSD(T) the Slater DZP basis is 40.7 mHa below cc-pVDZ at equal size, and 39.5 mHa below cc-pVDZ with its polarization re-optimized on the same footing. Of the gain, 28.8 mHa is at the Hartree–Fock level and 12 mHa is additional correlation energy. The evaluator is converged: going from twelve to sixteen Gaussians per function moves every energy by 0.07 mHa.

a. Place on the Gaussian ladder. The cc-pVTZ anchor makes the split readable. The Slater DZP RHF energy is 1.6 mHa above cc-pVTZ's, so it covers about 94% of the DZ-to-TZ mean-field gap. Its frozen-core CCSD(T) energy is 50.5 mHa above cc-pVTZ's, so it covers about 45% of the DZ-to-TZ correlated gap. The reading is the expected one: the exponential shape buys the mean-field part, the cusp at the nucleus and the exponential tail, while correlation energy is bought by the number of correlating functions, which is the same in both bases. That the mean-field advantage of Slater functions exists is textbook; what is measured here is its size at equal count under an identical design rule, and the fact that the correlated advantage is less than a third of it.

IV. THE FRAMEWORK'S OWN EXPONENTIAL KITS, NEGATIVE

The 41 mHa of Sec. III could come from the exponential shape or from the exponent tuning. The one-centre framework's radial rules fix the exponents, so they separate the two. Two rules and two remedies were tested, at the same twenty-four functions (shells $n \le 3$ on O, $n \le 2$ on H), the same geometry and the same solver, with no exponent or polarization tuned on water in any of them.

TABLE II. The exponential kits at 24 functions, frozen-core CCSD(T), same geometry and solver as Table I. STO-3G (7 functions) for reference. Every 24-function kit has 107 382 Jordan–Wigner Pauli terms on 46 qubits (Sec. VI); the frozen-core 1-norm is in hartree.

Kit	Fns	RHF	CCSD(T) fc	λ fc
STO-3G	7	-74.9629	-75.0123	
Sturmian, one k per atom	24	-74.2879	-74.5603	848.8
hydrogenic, $Z_{\text{eff}} = 6$ ($n \ge 2$)	24	-75.7466	-75.9471	405.1
hydrogenic, bare Z	24	-75.8397	-76.0452	510.8
Sturmian, two scales	24	-75.8903	-76.1113	620.0
cc-pVDZ	24	-76.0268	-76.2410	674.3
Slater DZP	24	-76.0556	-76.2817	706.3

Hydrogenic Z/n, bare Z.: Every function is a hydrogen-like orbital of the nuclear charge, decay Z/n, with its fixed radial polynomial. This is the production rule of the composed builder (Paper 17) and the two-centre engine (Paper 58). Zero parameters. A slow test checks that each function is a one-electron eigenfunction at $-Z^2/2n^2$ and that same-l functions are orthogonal; it caught two planted coefficient errors.

Hydrogenic with the composed screening.: O $1s$ at $Z = 8$; O shells $n \ge 2$ at $Z_{\text{eff}} = 6$, which is Z minus the $1s^2$ core as the composed classifier assigns it. Zero parameters. This maps the composed core and valence blocks onto one function set; the composed builder itself starts its valence block at $n = 1$, so the mapping is ours.

Coulomb–Sturmian, one scale per atom.: Avery's basis [1]: one shared exponent k on each centre. $k_{\text{H}} = 1$ makes the H $1s$ exact. $k_{\text{O}} = 3.781$ minimizes the radial oxygen energy, and the best that one shared scale can do for the atom is −71.406 Ha, 3.4 Ha above the Slater primitives' −74.768.

Coulomb–Sturmian, two scales.: A core $1s$ at k_c plus Sturmians sharing k_v (s with $n \le 2$, p with $n \le 3$, one $3d$) on oxygen, fourteen functions; $(k_c, k_v) = (7.793, 2.281)$ minimize the radial oxygen energy (−74.5027 Ha); H at $k = 1$. This is a simplified relative of Avery's generalized Sturmians, not his isoenergetic construction (Paper 60).

[MEASURED, negative] Table II gives the result. No kit with fixed or shared exponents reaches cc-pVDZ at twenty-four functions. The bare-Z hydrogenic kit is 196 mHa above it and 236 mHa above the Slater DZP. The one-scale Sturmian kit, at twenty-four functions, is worse than STO-3G at seven. So the 41 mHa of Sec. III came from exponent choice, not from the hydrogen-like shape.

a. Mechanism, hydrogenic. Bare $Z = 8$ puts the oxygen $n = 2$ shell at $\zeta = 4$, about twice as compact

as the tuned valence (1.45 to 2.85). The valence then has to live in the $n = 3$ functions ($\zeta = 2.67$), which leaves almost nothing to correlate with. The core is fine ($\zeta = 8$ against 7.7 to 9.3 tuned). Screening is the obvious remedy and it makes things worse, by 98 mHa. The diagnosis comes from the oxygen atom in the kits as built: bare −74.557, screened −74.420, and the same two kits with their polynomial contractions released give −74.757, an upper bound on what the contracted set could reach. So between 200 and 340 mHa is lost to the fixed hydrogenic node structure. Each hydrogenic function carries a radial polynomial whose nodes are set for orthogonality to the $1s$ of the *same* charge. Bare Z keeps every shell mutually consistent; $Z_{\text{eff}} = 6$ puts the valence nodes in the wrong place relative to the $Z = 8$ core. The contraction, not the decay rate, is the binding constraint.

b. Mechanism, Sturmian. One scale cannot serve a core near $\zeta = 8$ and a valence near $\zeta = 2$ at once. Sturmians pay no contraction penalty, since a shared-k set spans the same space as its primitives, and the two-scale kit is accordingly the best of the four, at 130 mHa above cc-pVDZ. The remedy the Sturmian literature names is several scales [1]; whether two is the cap or merely the first rung is what Sec. V tests.

c. An observation, not a claim. Across the twenty-four function kits other than the one-scale Sturmian, the frozen-core 1-norm rises with accuracy: 405, 511, 620, 674, 706 Ha. A low 1-norm at equal size here tracks a worse description, not a cheaper one. Sec. VI returns to this.

V. THE SCALE LADDER: HOW MANY OXYGEN EXPONENTS, SET ON THE ATOM

A. Design, fixed before any water rung was run

Twenty-four functions throughout. O: s primitives $(1s, 1s, 2s)$, p primitives $(2p, 3p)$, and one $3d$ tied to the $3p$ exponent. H: $1s, 2s, 2p$. Rungs differ only in which oxygen primitives share an exponent (Table III). Every oxygen exponent set minimizes the radial average-of-configuration Hartree–Fock energy of the free oxygen atom (Sec. II), Nelder–Mead from 41 starts. Hydrogen exponents are $k = 1$, except on the last rung, where $1s, 2s, 2p$ are each set by H_2 RHF at $R = 1.4$ bohr. Nothing is tuned on water. So any water gain is transfer, not fitting. The $R2$ rung is the two-scale Sturmian kit of Sec. IV; rebuilt through the ladder's driver it reproduces that kit's energies to 10^{-4} Ha (the pre-registered control SL-0).

The atom ladder is the control that frames the predictions. Its total gain $R2 \to R5$ is 257.3 mHa, and the p split ($R4 \to R5$), which releases the $2p$ exponent from the $3p$, carries 0.902 of it. Two transfer ratios were measured on cases the ladder does not score: the clean step from one shared oxygen scale to two (the two Sturmian kits of Sec. IV) transfers at 0.50, and the confounded step from the two-scale kit to the Slater DZP at 0.64, confounded upward because the DZP also has two hydrogen s exponents and water-tuned polarization.

TABLE III. The ladder. Letters mark which oxygen primitives share an exponent. Atom energies are the radial average-of-configuration Hartree–Fock minima that set the exponents; the seven-primitive Slater atom of Sec. III is −74.76820 Ha, 8.2 mHa below $R5$.

Rung	O groups $(1s, 1s, 2s, 2p, 3p)$	N_ζ	Atom	Step (mHa)
$R2$	a, b, b, b, b	2	-74.50266	
$R3$	a, b, b, c, c	3	-74.50605	3.4
$R4$	a, b, c, d, d	4	-74.52793	21.9
$R5$	a, b, c, d, e	5	-74.75997	232.0
$R5H$	as $R5$; H from H_2	5		

TABLE IV. Water on the ladder, 24 functions, same geometry and solver. Steps are rung-to-rung changes, positive downward. The atom step is the radial HF gain of Table III.

Rung	RHF	CCSD(T) fc	CCSD(T) ae	Water step	Atom step
$R2$	-75.890293	-76.111337	-76.120634		
$R3$	-75.890159	-76.108484	-76.117764	-2.85	3.4
$R4$	-75.905860	-76.124580	-76.137966	16.10	21.9
$R5$	-76.036106	-76.244112	-76.258281	119.53	232.0
$R5H$	-76.045000	-76.248015	-76.262063	3.90	

B. Predictions, as registered

The record [17] was committed on its own before any water energy of $R3$, $R4$, $R5$ or $R5H$ was computed. With $W(R)$ the frozen-core CCSD(T) water energy of rung R and $G = W(R2) - W(R5)$:

SL1 (gated).: $(W(R4) - W(R5))/G \geq 0.6$: the p split dominates on water too. Stated probability 0.75.

SL2 (gated).: $0.3 \leq G/257.3\,\text{mHa} \leq 0.8$, i.e. $W(R5)$ between −76.1885 and −76.3172 Ha. The band is wide on purpose, because the clean control measured a core step and the ladder's main step is valence. Probability 0.7.

SL3 (reported).: $W(R5)$ stays above the Slater DZP. Direction only. Probability 0.6.

SL4 (reported).: $W(R5H) < W(R5)$. Direction only. Probability 0.7.

Gate: GO if SL1 and SL2 both hold. Also reported, not predicted: $W(R5)$ against cc-pVDZ, which at the clean transfer ratio lands on top of it (−76.240 against −76.241) and so was declared a coin flip in advance.

C. Result

[**MEASURED, pre-registered; GO**] Table IV. $G = 132.78$ mHa.

- SL1 held: $(W(R4) - W(R5))/G = 0.900$, against the bound 0.6 and the atom's 0.902.
- SL2 held: $G/257.3\,\text{mHa} = 0.516$, inside the band; the clean control was 0.50.
- SL3 held in direction: $W(R5)$ is 37.7 mHa above the Slater DZP. The record's estimate from the confounded ratio had been about 6 mHa; that estimate was well off, and the record says so.
- SL4 held: $W(R5H)$ is 3.90 mHa below $W(R5)$, against an H_2 gain of 4.06 mHa, a transfer near 0.96 on the hydrogen side.

Reported, not predicted: against cc-pVDZ at the same twenty-four functions, $R5$ is 3.1 mHa below and $R5H$ is 7.0 mHa below. The two-scale kit had been 130 mHa above. So the gap of Sec. IV is closed by releasing exponents set on the free atom and on H_2, with nothing tuned on water. RHF carries most of it: 145.8 mHa from $R2$ to $R5$ at the mean-field level against 132.8 at CCSD(T).

What this does and does not say. $R5$ is a Slater-type basis with five atom-set oxygen exponents, a $3d$ tied to the $3p$, and one hydrogen exponent. It is a better-chosen member of the same primitive family as the two-scale kit, not a new structure. The Slater DZP, with water-tuned polarization and two hydrogen s exponents, stays 34 mHa below $R5H$. With the $3d$ also set on the atom, by its correlated energy, the gap is 21.7 mHa (Sec. V D); that is the price of setting everything on the atom by these rules. The lesson of Sec. IV stands: the case rests on one-centre structure, not on better bricks; what moved is that the two-exponent cap, not the Sturmian family, had been the 130 mHa.

D. The non-monotone rung, diagnosed

$R3$ splits the oxygen p exponent from the s-valence exponent. The atom gains 3.4 mHa; water loses 2.85 mHa. The rung-by-rung ratios of water step to atom step, −0.84, 0.735, 0.515, are therefore not one transfer number per ladder. A diagnostic, with its hypotheses written before any new number was computed, located the cause [17]. Nothing was tuned on water: every substituted exponent below is $R2$'s 2.28124 or $R3$'s own value.

The water loss is correlation. From $R2$ to $R3$, RHF moves +0.13 mHa, the CCSD correlation +2.57 and the triples +0.15. The one-centre probe is the neutral oxygen atom in the same Gaussian-fitted kits, with an unrestricted Hartree–Fock reference for the $M_S = 1$ triplet determinant and the same frozen-core CCSD(T). On the atom the step gains 2.84 mHa at UHF and loses 2.72 mHa of correlation, a wash of −0.12 mHa. Water loses the same 2.72 mHa of correlation. So the correlation half of the step transfers one to one, and the Hartree–Fock half does not transfer at all.

Where the correlation goes. The ladder's objective is Hartree–Fock on the atom, with no d function. It never sees the $3d$, and it sees the $3p$ only through the occupied $2p$. At $R3$ the p family drifts diffuse to suit the occupied $2p$, and the $3p$ and $3d$, which carry most of the correlation at this size and want to be compact, ride along. Putting the $3d$ alone back at $R2$'s exponent, with everything else at $R3$, recovers 2.13 of the 2.85 mHa on water; the atom had predicted about 2.1 before that run. About 80% of the loss is the $3d$, a passenger of the $3p$ by the ladder's design, with the rest split between the $3p$ and the s shift.

Why the Hartree–Fock gain cancels. The atom gains at HF from both moves of the step, the p family going diffuse and the s valence tightening. On water, the first costs 5.2 mHa at RHF and the second gains 5.1 mHa, so the atom's gain cancels to 0.13 mHa. Water's RHF is two to six times more sensitive than the atom's to the $3p$ and s-valence exponents, and of the opposite sign for the p family: the free atom's $2p^4$ wants a more diffuse p than the oxygen in water's bonds. The prediction's clause that RHF would stay within 0.2 mHa of $R3$ under each substitution failed for the $3p$ and s substitutions, by 6.7 and 5.1 mHa; the correlation clause held in all three. That failure is the mechanism.

The upshot for the ladder: $R3$'s ratio of −0.84 is a ratio of two cancelling HF pieces and a transferring correlation loss, not a transfer number, and the $R4$ and $R5$ rungs, which are HF-driven radial splits, are untouched. The ladder's one design flaw is identified: a $3d$ tied to an exponent the atomic objective cannot see.

[**MEASURED, pre-registered**] The fix was then run as its own registered ladder (record `docs/predictions/2026-11-01_scale_ladder_3d.md`, committed before any scored water energy). On every rung the $3d$ exponent is set by the free oxygen atom's frozen-core CCSD(T) energy, UHF triplet reference, with the s, p exponents unchanged; this is Dunning's correlation-consistent rule [3], and nothing is tuned on water. The atom puts the $3d$ at 2.74 to 2.80 on every rung, against 1.98 to 2.28 when tied to the $3p$. The prediction, from two control moves, was that water would change by the atom's correlated change within 15.6% on each rung. It held on all four: water over atom is 1.04, 1.05, 0.91 and 0.85 for $R3$, $R4$, $R5$ and $R5H$, the last at the band's edge. $R3$ stays 0.8 mHa above $R2$, as predicted: the $3d$ share of its loss is gone on both rungs, and what remains is the part that does not transfer. On the two largest moves the water RHF part rises instead of falling, so there the total is carried by correlation alone. The best atom-set rung is now $R5H$ at −76.260011 Ha, 19.0 mHa below cc-pVDZ and 21.7 mHa above the water-tuned Slater DZP.

TABLE V. Qubit resources, Jordan–Wigner. λ is the Pauli 1-norm in hartree, identity excluded. Frozen core (fc) drops the oxygen 1s.

Basis	Q fc	N_{Pauli} fc	λ fc	Q ae	N_{Pauli} ae	λ ae	CCSD(T) fc
Slater DZP	46	107 382	706.3	48	128 793	754.6	-76.2817
cc-pVDZ	46	107 382	674.3	48	128 793	717.0	-76.2410
cc-pVTZ	114	3 983 652	7 653.5	116	4 281 643	7 858.6	-76.3322

TABLE VI. Matched accuracy. Same geometry, solver and counter as Table V. "vs DZP" is the frozen-core CCSD(T) energy relative to the Slater DZP's, positive above.

Basis	Fns	CCSD(T) fc	vs DZP (mHa)	N_{Pauli} fc (Q)	λ f
cc-pVDZ	24	−76.2410	+40.7	107 382 (46)	674.
Slater DZP	24	−76.2817	0	107 382 (46)	706.
cc-pVDZ+TZpol	35	−76.2665	+15.2	505 467 (68)	1 814.
aug-cc-pVDZ	41	−76.2737	+8.0	982 785 (80)	1 901.
cc-pVTZ	58	−76.3322	−50.5	3 983 652 (114)	7 653.

VI. THE QUBIT COST

The second-quantized Hamiltonian of each basis was mapped to qubits by the Jordan–Wigner transformation [10] over RHF canonical orbitals, with interleaved spin, terms kept at $|c| > 10^{-10}$, the identity counted in the Pauli number and excluded from the 1-norm. The counter is closed-form per string class and is validated against a full Jordan–Wigner construction in OpenFermion on STO-3G water, where five planted errors were each caught; its frozen-core cc-pVDZ count matches the published count of Trenev *et al.* [11] exactly.

[**MEASURED**] Table V. Three facts.

1. The Pauli count is identical in every twenty-four-function basis, tuned or not, Gaussian or exponential. It is fixed by the orbital count and the C_{2v} pattern of the canonical orbitals, which every basis here shares, and not by the radial shape. Radial shape can move an energy by a hartree (Table II) without moving a single term.

2. The 1-norm is 4.7% higher for the Slater DZP than for cc-pVDZ. So the 41 mHa of Sec. III costs about five percent in 1-norm and nothing in qubits or terms.

3. cc-pVTZ, the next Gaussian rung, costs 37× the terms and 11× the 1-norm for the remaining 55% of the correlated gap.

The second fact is an equal-size statement, not a matched-accuracy one. The Slater DZP sits between cc-pVDZ and cc-pVTZ in energy, so the matched-accuracy question is which Gaussian basis first reaches its energy and what that basis costs.

A. Matched accuracy

Two intermediate Gaussian rungs were added to bracket the crossing from both sides, and the outcome was pre-registered before either was run [22]: cc-pVDZ's s and p shells with cc-pVTZ's polarization shells (O 2d, H 2p; 35 functions; our construction, there to bracket, not to recommend), and the standard aug-cc-pVDZ (41 functions). The gated predictions were that neither reaches the Slater DZP; the control bounds were measured on the neon atom first, where the second polarization set is worth 26 mHa, all correlation, and augmentation 30 mHa, a quarter of it mean-field.

[**MEASURED, pre-registered; GO**] Table VI. Both predictions held: the 35-function basis stays 15.2 mHa above the Slater DZP and aug-cc-pVDZ 8.0 mHa above it. Both steps are correlation-dominated, with mean-field shares of 0.19 and 0.45 against the Slater DZP step's 0.71, and the two rungs keep neon's order; those were the reported clauses, and they held too. The aug-cc-pVDZ energy agrees with the commonly quoted value near −76.274 Ha, which is an orientation check and not a cited reference.

The crossing therefore lies between aug-cc-pVDZ and cc-pVTZ, 0.14 of the way in energy. The lower bound needs no interpolation: aug-cc-pVDZ, still 8 mHa short of the Slater DZP, already costs 9.2× the Pauli terms, 2.7× the 1-norm, and 80 qubits against 46. Read linearly in energy between the bracketing rungs, the matched-accuracy Gaussian cost is near 1.4 million terms, a 1-norm near 2 700 Ha and about 85 qubits, which is 13 times the terms and 3.8 times the 1-norm; that interpolation is a reading of two measured rungs, not a measurement of any basis. The qubit statement of this paper is thus a matched-accuracy one: at matched or better accuracy on water, the Gaussian basis costs at least nine times the Pauli terms and 2.7 times the 1-norm of the Slater DZP, and the equal-size result of Table V is its lower rung.

Two things the angular sparsity of the one-centre framework [16] would have added, and does not here. Multi-centre integrals over these bases are dense; no Gaunt zero appears in any Hamiltonian of Table V, on either side. And the 1-norm observation of Sec. IV, that a low 1-norm at equal size tracks a worse energy, is the reason the composed library's 1-norm figures (Papers 14, 20) are reported at matched qubits with the accuracy stated alongside, never as a cost advantage on their own.

B. A second molecule: hydrogen fluoride

To test whether any of this is specific to water, the same rule was applied to hydrogen fluoride, the cheapest second molecule: a diatomic with ten electrons and 19 functions in cc-pVDZ, needing only two-centre integrals. Nothing in the rule changed. The fluorine

TABLE VII. Hydrogen fluoride at $r_e = 1.7325$ bohr, F $1s$ frozen, same solver and counter as Table VI. "vs DZP" is relative to the Slater DZP, positive above. cc-pVDZ*: its F d and H p exponents re-optimized on the molecule, as for the Slater basis.

Basis	Fns	CCSD(T) fc	vs DZP (mHa)	N_{Pauli} fc (Q)	λ fc
cc-pVDZ	19	−100.2281	+59.6	63 535 (36)	433.6
cc-pVDZ*	19	−100.2282	+59.5	63 535 (36)	434.8
cc-pVDZ+TZpol	27	−100.2551	+32.6	290 731 (52)	1 089.9
aug-cc-pVDZ	32	−100.2636	+24.1	591 674 (62)	1 169.9
Slater DZP	19	−100.2877	0	63 535 (36)	432.6
cc-pVTZ	44	−100.3384	−50.6	2 096 366 (86)	4 483.5

TABLE VIII. H_6 ring, D_{6h}, 1.0 Å, all electrons, CCSD(T). cc-pVDZ*: its p exponent re-optimized on the ring. "vs DZP" is relative to the Slater DZP, positive above. Pauli counts over RHF canonical orbitals, tolerance 10^{-10}.

Basis	Fns	CCSD(T)	vs DZP (mHa)	N_{Pauli} (Q)	λ
cc-pVDZ	30	−3.387147	+4.8	286 927 (60)	987
cc-pVDZ*	30	−3.388492	+3.5	284 895 (60)	944
Slater DZP	30	−3.391958	0	288 959 (60)	910
aug-cc-pVDZ	54	−3.393029	−1.1	3 038 807 (108)	2 846
cc-pVDZ+TZpol	48	−3.395593	−3.6	1 812 729 (96)	3 290
cc-pVTZ	84	−3.409413	−17.5	16 902 041 (168)	14 679

primitives were set on the free atom's radial average-of-configuration Hartree–Fock energy and contracted as cc-pVDZ is; the hydrogen exponents are water's; the polarization exponents of both the Slater basis and cc-pVDZ were re-optimized on the molecule's frozen-core CCSD. The predictions and their bounds were registered first [23], with neon, the isoelectronic closed shell, as the measured control: there the Slater DZP is 81.4 mHa below cc-pVDZ at fourteen functions, an upper anchor because a free atom's own contracted orbitals sit near its Hartree–Fock limit.

[MEASURED, pre-registered; GO] Table VII. All five gated or scored predictions held. At equal size the Slater DZP is 59.5 mHa below the re-optimized cc-pVDZ, between water's gap and neon's as predicted, and 0.69 of that is mean-field, as for water (0.71) and neon (0.70). The Pauli count is identical, 63 535 terms on 36 qubits for both, and here the Slater 1-norm is not higher but 0.2% lower. At matched accuracy the pattern repeats: aug-cc-pVDZ at 32 functions is still 24.1 mHa short, the crossing lies between it and cc-pVTZ (0.32 of the way, against water's 0.14), and aug-cc-pVDZ already costs 9.3× the Pauli terms and 2.7× the 1-norm on 62 qubits against 36. The two molecules give nearly the same two ratios; that was reported, not predicted, and is not read as a law.

C. A third system: a ring of six hydrogens

Both molecules above have a heavy atom. To see whether the result needs one, the same rule was applied to H_6, a regular hexagon at 1.0 Å, with no core at all [24]. The Slater DZP is $[2s1p]$ per hydrogen with water's $1s$ pair; its $2p$ exponent and cc-pVDZ's were re-optimized on the ring's CCSD. All electrons are correlated. The solver is the same, through a memory-lean copy that never forms the spin-orbital virtual block; it gives the original's energies to every printed digit on three smaller rings. On the scored ring it reproduces the published cc-pVTZ CCSD, −3.405645 Ha [25], to 4×10^{-7} Ha. The predictions were scaled from H_2: three H_2 gaps make 12.3 mHa.

[MEASURED, pre-registered; STOP] Table VIII. At equal size the Slater DZP still wins, but by 3.5 mHa, short of the registered bar of 6.2 mHa. The mean-field part is 4.9 mHa, larger than the whole gap: correlation gives 1.5 mHa back. At matched accuracy the result turns over. One more Gaussian polarization shell, cc-pVDZ+TZpol at 48 functions, is already 3.6 mHa below the Slater DZP, and aug-cc-pVDZ is 1.1 mHa below. So the crossing lies between 30 and 48 functions, and the cheapest Gaussian basis that beats the Slater DZP costs 6.3× the Pauli terms and 3.6× the 1-norm. The water and hydrogen fluoride margins at matched accuracy are not a property of exponential functions in general; on this ring they shrink to one shell.

The ring also qualifies the Pauli-count statement. Over canonical orbitals the three 30-function counts differ by up to 1.4%, even between the two cc-pVDZ forms. The cause is the ring's degenerate orbital pairs: the solver returns an arbitrary rotation inside each pair, and that moves integrals across the tolerance. In a fixed symmetry-adapted labelling the counts are identical for the Slater and Gaussian bases: 149 591 in ring labels and 653 971 in site labels, the two labellings described next. Water and hydrogen fluoride have no degenerate orbitals, which is why their canonical counts matched. The registered prediction of equal canonical counts missed.

Two labellings of the same orbital space were also compared, registered in the same record. *Site* labels are orthogonalized local functions; *ring* labels are their discrete Fourier transform around the ring. Ring labels carry the ring momentum, which gives two more qubit symmetries (five against three) and 4.37 times fewer Pauli terms at 10^{-10}, since 87% of their integrals vanish exactly. Site labels give fewer terms once small ones are dropped: at 10^{-2}, 4 267 against 7 987. The crossover falls between 10^{-4} and 10^{-3}. All three predictions held in both the Slater and the cc-pVDZ space. Using symmetries to remove qubits is standard [26]; that local orbitals suit some methods and delocalized ones others is reported for hydrogen chains, rings and lattices [27]; the counts here are new only as measurements. A comparison of the two families on hydrogen rings at the Hartree–Fock level, with one function per atom, is Ref. [28].

[MEASURED, pre-registered; GO] A torus of rings has two winding labels instead of one [29]. The molecule is N small rings of M hydrogens, $M = 3$ or 4,

set upright around a circle, with one minimal Slater 1s per atom. The winding k around the hole is exact, since every ring is equivalent under the N-fold rotation. The winding m around the tube is not: the inner side of the tube faces the hole and the outer side does not. Eleven scored cases were run after sixteen controls at a different ring spacing set every band. The share of the integrals that breaks k is at most 8.4×10^{-15}. The qubit symmetries count 2, 4 and 5 in site labels, k labels and (k, m) labels at even N, and 2, 3 and 4 at odd N, as counted in advance. The share that breaks m falls as the torus thins. Its local slope against the ratio r/R of tube radius to ring radius rises toward 1 for a change of m by one (0.943 at $M = 3$ and 0.998 at $M = 4$, in the electron-pair integrals at the thinnest step) and toward 2 for a change by two (1.980 at $M = 4$). That is the curvature coupling of toroidal and poloidal modes known for electrons on a torus [30]; here it is checked in a molecule. Using m as well as k always keeps the same fraction of the Pauli terms at 10^{-10}: 0.5065 at $M = 3$ and 0.5308 at $M = 4$, unmoved by the change of spacing. That fraction has no derivation and is reported as an observation. At 10^{-2} site labels again give the fewest terms. Momentum labels for periodic systems and their use in removing qubits are established [31, 32].

VII. PRIOR ART

A moderate search was made, against primary sources where reachable. The mean-field half of Sec. III, that Slater functions beat Gaussians per function at the SCF level, is textbook and no part of it is claimed here. The design recipe, atom-optimized Slater primitives generally contracted on the cc-pVXZ pattern, has precedent in the Madrid group's STO valence sets for H to Ne, a single d for B to Ne, with all exponents set by minimizing the sum of the atom's Hartree–Fock and frozen-core CISD energies and hydrogen's on H_2 [12], and in the correlation-consistent-style STO sets of Förster and Visscher [13], and earlier in the correlation-consistent Slater sets of Lesiuk and co-workers for the beryllium dimer, who report at FCI and CCSD(T) that they "have not found a situation when STOs perform worse than GTOs basis sets of the same size" [19]; that is the qualitative form of the equal-size result here, and it is theirs. The counterpoint is also on record: stock Slater sets tuned for density-functional work need about 45% more functions than 6-311G for the same Hartree–Fock precision [20], which fits this paper's reading that the gain is in how the exponents are designed, not in the function family. The closest molecular STO-against-cc-pVXZ comparison found compares forces, not correlated energies [14]. No quantitative matched-size or matched-accuracy table of correlated energies on a polyatomic molecule was found, nor the split of the DZ-to-TZ gap into its mean-field and correlated parts, nor any qubit-resource comparison between the two families. The statement that the Pauli count is fixed by orbital count and symmetry holds for the delocalized canonical orbitals used here; it is not a general law, since localized or diagonal bases change the count [21], and degenerate orbitals make the canonical count depend on an arbitrary rotation (Sec. VI C). A qualitative reading that orbital shape matters little for correlation energy is sometimes attributed to Persson and Taylor; the full text could not be read and it is not cited as support. The water complete-basis anchor of Helgaker, Klopper, Koch and Noga [15] was consulted for orientation only; no basis-set-limit statement is made.

The transfer question of Sec. V, whether exponents set on the free atom carry to the molecule and at what ratio, is the standard premise of every atom-optimized basis family, cc-pVXZ included [3]. What is specific here is the pre-registered rung-by-rung measurement of the ratio, the identification of the p-split as the carrier on both sides, and the diagnosis of the one rung where the ratio has the wrong sign.

VIII. LIMITATIONS

- One geometry and one method per system, and three systems. The equal-size and matched-accuracy findings are checked on hydrogen fluoride (Sec. VI B), and they fail on the H_6 ring (Sec. VI C): the equal-size gap shrinks to 3.5 mHa and the matched-accuracy margin to one shell. The native kits, the ladder and the $R3$ diagnosis are water only. No potential energy surface, no basis-set limit. The ladder frees exponents in one order and does not test every partition of five primitives. "Needed" in Sec. V means "worth more than the bound at twenty-four functions"; it does not mean a count is forced.

- The Slater functions are Gaussian fits. The fit is converged at the 0.07 mHa level for the DZP basis and is exponent-independent by construction on the ladder, but every number here is for the fitted basis. Exact Slater integrals on three centres exist in the series (Paper 59) and were not used for these energies.

- The matched-accuracy comparison (Sec. VI A) uses standard Gaussian bases as published against the Slater DZP as built, with its water-optimized polarization; cc-pVDZ's own re-optimized polarization is worth 1.2 mHa and does not change the bracket. The 35-function intermediate basis is our construction. The interpolated matched-accuracy cost is a reading between two measured rungs.

- The 3d exponent on the ladder is tied to the 3p, and Sec. V D shows that tie to be the ladder's one design flaw. Setting the 3d by an atomic correlated criterion is Dunning's correlation-consistent rule,

correlating exponents "optimized in atomic correlated calculations" while the s, p functions "can be taken from atomic Hartree–Fock calculations" [3]; a ladder built that way is a different ladder; it was registered separately and run (Sec. V D).

- The solver is the project's own. It is validated against an external reference on STO-3G water and against exact limits. The one larger-basis external check is of the correlation energy alone, on hydrogen fluoride in another group's Slater basis (Sec. II); no external total energy in these bases exists on the host.
- None of this is a GeoVac lattice result, and none of it uses the angular sparsity that is the framework's structural asset. The paper measures what tuned exponential functions buy at equal size, and finds that the framework's fixed-exponent rules do not buy it.

IX. CONCLUSION

At twenty-four functions on water, a Slater-type basis built by cc-pVDZ's own rule is 41 mHa lower at frozen-core CCSD(T), mostly at the mean-field level, for the same Pauli count and five percent more 1-norm. The gain is exponent choice: the one-centre framework's fixed-exponent rules, hydrogenic Z/n and single-scale Sturmian, lose by 0.2 to more than 1 Ha at the same size, and the hydrogenic loss is the node structure the fixed exponents impose, which screening makes worse. Releasing the oxygen exponents one at a time, each set on the free atom, closes that gap: five exponents put water 3 mHa below cc-pVDZ with nothing tuned on the molecule, the p split carries nine tenths of the gain on both the atom and the molecule, and the gain transfers at about half, all as predicted before the energies were computed. The one rung that went the wrong way is a $3d$ passenger the atomic objective cannot see, plus two mean-field pieces that cancel on water and not on the atom. At matched accuracy the comparison turns around: aug-cc-pVDZ at 41 functions is still 8 mHa short of the Slater DZP and already costs nine times the Pauli terms and 2.7 times the 1-norm, and the crossing lies between it and cc-pVTZ. Hydrogen fluoride, built by the same rule and registered in advance, repeats both the equal-size gap (59.5 mHa, mostly mean-field, same Pauli count) and the matched-accuracy bracket. A ring of six hydrogens, with no core, does not: there the Slater DZP is 3.5 mHa ahead at equal size and one extra Gaussian shell overtakes it, so the matched-accuracy margin is not general to exponential functions. Setting the $3d$ on the atom's correlated energy too, Dunning's rule, run as its own registered ladder, puts water 19.0 mHa below cc-pVDZ, with every rung's change predicted from the atom to within 15%. The water-tuned basis stays 21.7 mHa ahead of that best atom-set rung, and nothing here is a property of the lattice.

BACKING

Solver, validation tests, the radial atom, the kit builders and the Pauli counter: `tests/test_water_ccsd.py`. The ladder's guard, `tests/test_paper62_scale_ladder.py`, recomputes the free-atom objective at the stored exponents and checks that each rung's exponents are a stationary minimum of it, derives the SL1 and SL2 ratios from the stored water energies and the recomputed atom energies, and recomputes the one-centre $R3$ probe of Sec. V D; only the water energies are read from its data file. The matched-accuracy guard, `tests/test_paper62_matched_accuracy.py`, recomputes the aug-cc-pVDZ and cc-pVDZ water runs end to end, energies and Jordan–Wigner cost, from basis files stored beside it; the Slater DZP target and the other two rungs are read as registered values. The ladder's record, with its predictions, disposition and the $R3$ addendum: `docs/predictions/2026-10-30_scale_ladder.md`; the matched-accuracy record: `docs/predictions/2026-10-31_matched_accuracy_gaussian.md`; the hydrogen fluoride record: `docs/predictions/2026-10-31_hf_slater_second_molecule.md`. The hydrogen fluoride guard, `tests/test_paper62_hf_second_molecule.py`, recomputes the free fluorine atom's design step and the cc-pVDZ and aug-cc-pVDZ runs end to end; the Slater DZP and the other two rungs are read as registered values. The hydrogen ring and hydrogen torus records, each with its controls and disposition: `docs/predictions/2026-11-01_hydrogen_ring.md` and `docs/predictions/2026-11-02_hydrogen_torus.md`. Chronicle: CHANGELOG v5.20.11, v5.20.12, v5.21.25, v5.21.28, v5.21.29, v5.22.1.

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